

TrES-5b

R-1138

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_190504 · created 2026-07-30T23:44:32+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458607.8448 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.01
Transit depth (Rp/Rs)^2	3.4 +/- 0.38 [%]
Orbital Inclination (inc)	89.19 +/- 1.33
Airmass coefficient 1 (a1)	1.237 +/- 0.042
Airmass coefficient 2 (a2)	-0.1121 +/- 0.007
Scatter in the residuals of the lightcurve fit is	3.33 %
Best Comparison Star	None
Optimal Aperture	2.86
Optimal Annulus	5.93
Transit Duration (day)	0.09 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.01 vs 0.142 (deviation 2.08 σ)
Duration fit vs published	0.09 d vs 0.0758 d (deviation 4.67 σ)
Mid-time O-C	-7.1 min
Depth SNR	9.1
Residual scatter	3.33 %

Light curve

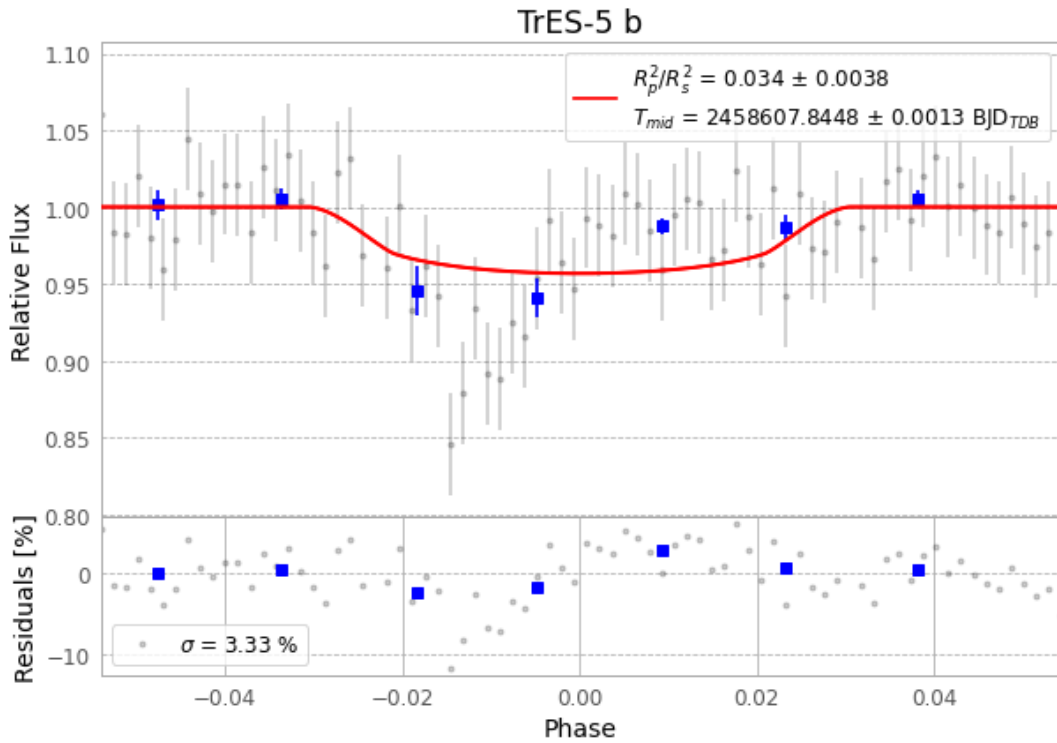


Figure 1 — FinalLightCurve_TrES-5 b_2019-05-03.png (SHA-256 242fecdd98bd4166...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 93], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ff50b47e4079b8d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), TRES-5190504062112.FITS → TRES-5190504101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-190504.log (4a77f7cd5f41...), FinalLightCurve_TrES-5 b_2019-05-03.png (242fecdd98bd...), FinalLightCurve_TrES-5 b_2019-05-03.csv (cb83eaccf767...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:44Z.